Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. TEAM AND PROJECT OVERVIEW

Team Name

KMJDIHATIKU

Selected Domain

Domain 1: The Aiversecity Future-Ready Workforce

Selected SDG Track (1–17)

SDG 4 (Quality Education), also SDG 8

Project / Solution Name

BijakMatch (ReRole Green)

Name (Leader)

Chong Poh Yi

Email & Phone (Leader)

University Affiliation(s)

University Malaya

Number of Team Members

2

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Malaysia is changing faster than its workers can retrain. TalentCorp's 2024 study found about 620,000 jobs in ten key sectors, 18% of the 3.5 million workers there, highly impacted by AI, digitalisation and the green economy. The Ministry of Science, Technology and Innovation says more than 600,000 workers need reskilling within three to five years, and only 26% of Malaysian professionals have formal AI training (Hays Asia Salary Guide 2025).

TVET workers are hit hardest: about 439,000 are in TVET today (Deputy Prime Minister, Oct 2025), plus many working technicians. A CV says little about what they can do. "Technician, 5 years" hides the faults they fix, the certificates they hold and the evidence they have. AI news is told as "your job will be replaced", not as which tasks are changing. And "go green" has no first step: workers cannot tell whether a skill is a lasting asset or a fad, whether they can learn it in time, or which statutory licence they still lack.

This matters for decent work and productivity (SDG 4.3, 4.4, 8.2, 8.5). Our sample worker, Encik Ahmad, is a 28-year-old E&E production technician with 4 free hours a week and RM100 a month. A degree or a generic AI course does not fit him. He needs to know which of his current tasks carry over, whether the target job is already overcrowded, and what to do first.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

BijakMatch is a career planner for TVET workers. It reads what a worker can really do, shows how AI and the green economy change those tasks, and turns the best next step into a weekly study plan.

1. Evidence beyond the CV. Workers add certificates (SVM, SKM), a daily work log and a short voice or typed shift interview. A language model asks the questions, but every answer must map to a Malaysian NOSS task card or be marked unclassified. Skills are sorted into Verified, Hidden (done at work, no certificate) and Rusty.
2. AI-trend signal layer. Dated snapshots of the Anthropic Economic Index and the ILO generative-AI exposure index set a moderate and an accelerated scenario. Labour-market signals come from OpenDOSM, MyMAHIR, MGTC and HRD Corp. Every source shows its date.
3. Explainable matching. Stay, Evolve and Pivot are scored by fixed, published rules (task overlap, demand, interest, ease of move, SDG bonus) plus a Window Score. Scoring is deterministic and never uses a generative model. Market crowding is shown next to each path, and each path links to the SDGs.
4. Adaptive pathway. Courses are ordered (recognise prior learning, refresh rusty skills, close gaps, then licences) and placed into a calendar. Missing a week or switching path rebuilds the plan.

Workshop skills used: GenAI APIs (Claude Opus 5.5, GLM-5.3) with schema-checked output, a Node.js backend, and cloud hosting (Vercel for the web app, RunPod for GPU speech-to-text and model training, LiveKit for live voice).

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Front end: HTML, CSS and plain JavaScript (no framework), hosted on Vercel. Back end: Node.js 24 server with the Anthropic SDK, LiveKit server SDK and Zod schema validation.
AI: Claude Opus 5.5 (shift interview, live market report with web search); GLM-5.3 on RunPod (interview fallback); Whisper large-v3 via faster-whisper on a RunPod Serverless GPU (Bahasa Malaysia and English speech-to-text); an xlm-roberta-base task-card classifier fine-tuned on a RunPod Pod; a LiveKit Cloud voice agent for the live skill check.
Data: versioned snapshots of the Anthropic Economic Index, ILO exposure index, OpenDOSM, MyMAHIR and HRD Corp (no database needed). All scoring weights live in one file, shared/rules.v1.js (v1.0.0).

Block diagram:
Worker (browser) -> Node API
  /api/interview -> Claude or GLM-5.3 -> schema check -> NOSS task cards
  /api/transcribe -> RunPod Whisper -> text
  /api/market -> Claude + web search -> sourced report
  /api/livekit/token -> LiveKit voice agent
Task cards + AI-trend and market snapshots -> rule engine -> Stay / Evolve / Pivot, Window Score, weekly planner -> study calendar and Career Transition Card.

Official SDC sponsor resources: YES. RunPod: GLM-5.3 public endpoint, a Serverless GPU endpoint (Whisper) and a GPU Pod (classifier training). Replit: development environment only.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we studied the TalentCorp, TVET and Hays figures and built Encik Ahmad from the proposal as a persona (synthetic: 4 hours a week, RM100 a month). Define: three pains, the CV hides skills, AI news scares instead of guiding, and "go green" has no first step. Ideate: a three-screen flow, Understand, Decide, Act, where each screen answers one question. Prototype: a working web app with a guided demo tour of the four core features. Test: we reviewed the interface through screenshots and recorded walkthroughs, simplified it into a static layout with plain wording, and run automated tests (npm test) covering scoring, the RunPod client and the backend end to end, including offline fallbacks.

Deployment: the static front end is on Vercel and the Node API runs separately. Secrets stay in environment variables, never in the repository, and every AI feature has a fallback if a key or service is missing. Next, by Phase 2: pilot with a TVET provider, turn market crowding into a scored rule, add a fairness test (changing age, gender or background must not change a recommendation), and add more states and jobs.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4.4: increase the number of people with relevant technical and vocational skills for decent jobs. SDG 4.3: equal access to affordable technical and vocational education and lifelong learning. Supporting: SDG 8.5 (full and productive employment) and SDG 8.2 (productivity through innovation).
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	A TVET worker's CV shows only part of their real skills. In our synthetic sample worker (Encik Ahmad), only 4 of 10 real skills (40%) show on paper; 6 are hidden or unrecorded. Context: 56% of Malaysian professionals use AI at work but only 26% have formal training (Hays Asia Salary Guide 2025). About 620,000 jobs in 10 sectors are highly impacted (TalentCorp, Nov 2024). More than 600,000 workers need reskilling within 3 to 5 years (MOSTI, quoted in the SDC Domain 1 brief).

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	% of a worker's real skills recognised on their profile = recognised skills / all skills found (Verified + Hidden + Rusty), where a skill counts as recognised when it has evidence and a route to certification (for example PPT). Secondary: weeks to a first concrete step (target: within 4 weeks).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 80% of real skills recognised by the final prototype presentation, up from 40% (4 of 10) in the baseline sample, measured on the sample worker profiles.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

N/A. Everything in this submission (code, design, research, proposal) was created during the SDC 2026 hackathon period.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Whisper large-v3 (via faster-whisper)	model (pre-trained)	MIT
xlm-roberta-base (we fine-tuned it)	model (pre-trained base)	MIT
Anthropic Economic Index; ILO GenAI exposure index; OpenDOSM	dataset (public)	CC-BY-4.0; ILO terms; OpenDOSM open data
@anthropic-ai/sdk, livekit-server-sdk, zod	code (open source)	Apache-2.0; Apache-2.0; MIT
GSAP 3.12.5 and Motion 11.11.13 (CDN)	code (UI libraries)	GSAP Standard licence; MIT

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

N/A. No team member is bound by an employment, internship or consulting agreement that could claim this work.

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

[Date]